

# Inference and Machine Learning

Part 1: How do we represent information and uncertainty mathematically?

## Probabilistic Machine Learning

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August 27, 2026

# Last time, and today

## Last time

Probabilistic reasoning with known distribution: Five numbers specified the model (traffic, Alice, Bob), and probability alone answered every question we asked.

## Today

The distributions are unknown. We start from data instead, in two problem types: **inference** and **supervised learning**.

**Keep asking:** what is unknown, and where does the uncertainty come from?

# Outline

- 1 Inference
- 2 Supervised machine learning
- 3 What knowing the joint would buy

# A question about a gene

We are investigating a specific allele of a gene: how likely is a carrier to develop a certain disease?

## The unknown

$\theta$ : the probability that someone carrying the allele develops the disease.

## The data

Among 100 people with the allele, 2 have the disease.

An answer is a number for  $\theta$ , together with a statement of how sure we are about it.

**Question:** what is  $\theta$ ?

## Two views of $\theta$

### Frequentist

$\theta$  is fixed and unknown. We estimate it, and ask how good the estimation procedure is.

### Bayesian

$\theta$  is a random variable: we could be living in a universe where  $\theta$  is 0.01, or 0.05, or any other value, and in many cases we will never find out which. We describe our belief with a prior, and update it with the data.

The same data and the same question, but two different notions of what “unknown” means. In the Bayesian view, taking  $\theta$  as fixed would be a contradiction.

# The frequentist view

## Estimators

The frequentist framework does not prescribe a single way to estimate  $\theta$ : many **estimators** are possible. A natural option is the sample mean, the fraction of the sample with the disease:  $\hat{\theta} = 2/100 = 0.02$ .

## How accurate is our estimate?

We cannot know: judging  $\hat{\theta} = 0.02$  would require knowing the true  $\theta$ .

## How accurate is the estimator?

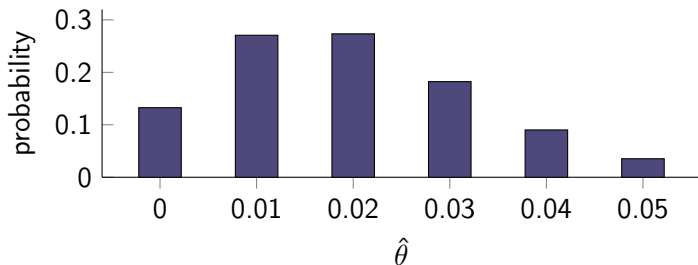
What we can ask is how accurate the *estimator* is, and that depends on how well the data represents the true value, and how well the method extracts information from it.

Accuracy is a property of the procedure, evaluated over repeated sampling.

## How accurate is the estimator?

We study how the estimator behaves under all possible data samples.

If  $\theta$  were really 0.02, the count of diseased people in a sample of 100 is  $\text{Bin}(100, 0.02)$ , and  $\hat{\theta} = \text{count}/100$ ; each sample gives a different count. The resulting distribution of  $\hat{\theta}$ :



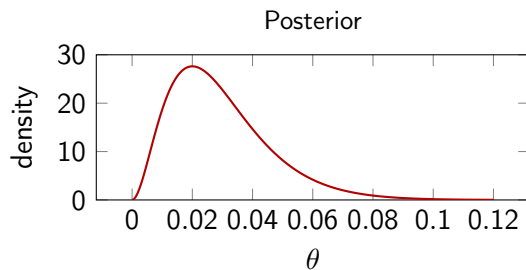
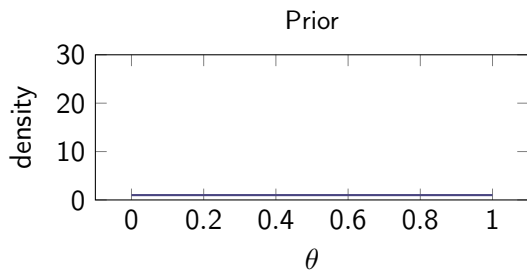
The estimate is 0.02 only 27% of the time, and the remaining mass above 0.05 is about 0.016.

**Later:** quantifying this spread precisely, and the apparent contradiction of assuming we know  $\theta$ , are the subject of the frequentist estimation chapter.

# The Bayesian view

## Prior and posterior

Prior: uniform on  $[0, 1]$ . Later in the course we will be able to calculate the posterior  $p(\theta \mid \text{data})$  from the prior and the data; for 2 of 100, it is plotted below.



The posterior is peaked at  $\theta = 0.02$  (its mode), with mean 0.029, but still allows other values.

The posterior is the whole answer: a number for  $\theta$  (its mode, 0.02) and how sure we are (its spread), in one distribution.

## Source of uncertainty: finite sample size

In both views, the uncertainty comes from having only a finite sample.

### More data, less uncertainty

With the disease status of 20 out of 1000 carriers (the same rate, ten times the data), the distribution of the estimate  $\hat{\theta}$  and the posterior of  $\theta$  are both more concentrated: lower variance.

Inference uncertainty shrinks as the sample grows: with more carriers, the estimate tightens, in both views.

# Think, pair, share

## Who can say it: a frequentist, a Bayesian, both, or neither?

- (a) The count is a random variable, unobserved before the study.
- (b) Before the data, every value of  $\theta$  in  $[0, 1]$  was equally plausible.
- (c) After the data, the probability that  $\theta$  lies between 0.01 and 0.03 is about 0.5.
- (d) Our estimate, 0.02, is within 0.01 of the true  $\theta$ .
- (e) Our estimator is within 0.05 of the truth at least 68% of the time, whatever the true  $\theta$ .
- (f) To evaluate the estimator, we need to consider even samples we did not see.
- (g) A study of 1000 carriers would leave less uncertainty about  $\theta$  than ours.

## True or false?

- (h) An accurate supervised model with additional features for predicting disease onset would necessarily give us an accurate  $\theta$  too.
- (i) The Bayesian answer is more accurate than the frequentist one.

**Think** alone. **Pair**: settle every disagreement with your neighbour. **Share**: with the class; write down some of your conclusions, with both names, and hand it in.

# Think, pair, share: answers

## Who can say it, and why

| Statement                                                          | Who | Why |
|--------------------------------------------------------------------|-----|-----|
| (a) The count is a random variable                                 |     |     |
| (b) Every $\theta$ equally plausible at first                      |     |     |
| (c) $\Pr(0.01 \leq \theta \leq 0.03 \mid \text{data}) \approx 0.5$ |     |     |
| (d) The estimate is within 0.01 of $\theta$                        |     |     |
| (e) Within 0.05 of $\theta$ at least 68% of the time               |     |     |
| (f) Evaluating the estimator needs unseen samples                  |     |     |
| (g) 1000 carriers, less uncertainty                                |     |     |

## True or false, and why

(h)

(i)

# Think, pair, share: answers

## Who can say it, and why

| Statement                                                          | Who  | Why                                 |
|--------------------------------------------------------------------|------|-------------------------------------|
| (a) The count is a random variable                                 | Both | modelled the same way in both views |
| (b) Every $\theta$ equally plausible at first                      |      |                                     |
| (c) $\Pr(0.01 \leq \theta \leq 0.03 \mid \text{data}) \approx 0.5$ |      |                                     |
| (d) The estimate is within 0.01 of $\theta$                        |      |                                     |
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| (g) 1000 carriers, less uncertainty                                |      |                                     |

## True or false, and why

(h)

(i)

# Think, pair, share: answers

## Who can say it, and why

| Statement                                                          | Who              | Why                                              |
|--------------------------------------------------------------------|------------------|--------------------------------------------------|
| (a) The count is a random variable                                 | Both<br>Bayesian | modelled the same way in both views<br>the prior |
| (b) Every $\theta$ equally plausible at first                      |                  |                                                  |
| (c) $\Pr(0.01 \leq \theta \leq 0.03 \mid \text{data}) \approx 0.5$ |                  |                                                  |
| (d) The estimate is within 0.01 of $\theta$                        |                  |                                                  |
| (e) Within 0.05 of $\theta$ at least 68% of the time               |                  |                                                  |
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|--------------------------------------------------------------------|----------|-------------------------------------|
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| (b) Every $\theta$ equally plausible at first                      | Bayesian | the prior                           |
| (c) $\Pr(0.01 \leq \theta \leq 0.03 \mid \text{data}) \approx 0.5$ | Bayesian | the posterior gives 0.51            |
| (d) The estimate is within 0.01 of $\theta$                        |          |                                     |
| (e) Within 0.05 of $\theta$ at least 68% of the time               |          |                                     |
| (f) Evaluating the estimator needs unseen samples                  |          |                                     |
| (g) 1000 carriers, less uncertainty                                |          |                                     |

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| (e) Within 0.05 of $\theta$ at least 68% of the time               |          |                                      |
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| (e) Within 0.05 of $\theta$ at least 68% of the time               | Frequentist | the procedure's worst case over $\theta$ |
| (f) Evaluating the estimator needs unseen samples                  |             |                                          |
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| (f) Evaluating the estimator needs unseen samples                  | Frequentist | judged over all possible samples         |
| (g) 1000 carriers, less uncertainty                                |             |                                          |

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| (g) 1000 carriers, less uncertainty                                | Both        | finite sample size, in both views        |

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## True or false, and why

- (h) False: the model gives  $p(y \mid x)$  per feature vector;  $\theta$  averages it over carriers, a separate estimation step.
- (i)

# Think, pair, share: answers

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| (g) 1000 carriers, less uncertainty                                | Both        | finite sample size, in both views        |

## True or false, and why

- (h) False: the model gives  $p(y \mid x)$  per feature vector;  $\theta$  averages it over carriers, a separate estimation step.
- (i) False: same target  $\theta$ . Frequentist: no prior to choose, but no automatic recipe for an estimator. Bayesian: the posterior follows mechanically from prior and data, but depends on the prior.

# Outline

- 1 Inference
- 2 Supervised machine learning
- 3 What knowing the joint would buy

# The house-price problem

Predict the sale price of a house from its location, lot size, square footage, and number of bedrooms.

## Data

$$\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}, \quad \mathbf{x}_i \in \mathcal{X}, y_i \in \mathcal{Y}.$$

$\mathcal{X}$  is the feature space,  $\mathcal{Y}$  the label space: here  $\mathbf{x}_i = (\text{location, lot size, square footage, bedrooms})$  and  $y_i = \text{the sale price}$ .  $f(\mathbf{x})$  is a prediction of  $y$  – similar to what Zillow does.

We are not told the rule relating features to price, only examples of it.

# Goal and evaluation

## Predictor

A function  $f$  with  $\hat{y} = f(\mathbf{x})$ .

## Loss and risk

Loss on one point:  $L(y, f(\mathbf{x}))$ : absolute  $|y - f(\mathbf{x})|$ , squared  $(y - f(\mathbf{x}))^2$ , or 0-1 (correct/incorrect), depending on the problem. **Population risk**, the expectation under  $p(\mathbf{x}, y)$ :

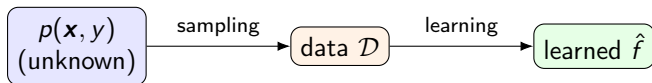
$$\mathcal{L}(f) = \mathbb{E}[L(Y, f(\mathbf{X}))].$$

## The learning problem

$$f^* = \arg \min_f \mathcal{L}(f)$$

# The catch

We do not have  $p(\mathbf{x}, y)$ . We have  $\mathcal{D}$ .



Two routes (both later in the course)

- Minimize the **empirical risk** on  $\mathcal{D}$  directly.
- Estimate  $p(\mathbf{x}, y)$  from  $\mathcal{D}$ , then predict with the estimate.

## Sources of uncertainty: both

### Noise

Two houses identical on paper can sell for different amounts: factors left out of the model.

### Finite sample size

The relationship between features and price is learned from a finite record of past sales.

Supervised learning combines both sources of uncertainty that reasoning and inference each had alone.

**Later:** Sep 1 shows how quickly finite data runs out when a language model is learned by counting.

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- 1 Inference
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If  $p(\mathbf{x}, y)$  were known, the problem would be solved

### Regression: squared loss

$L(y, f(\mathbf{x})) = (y - f(\mathbf{x}))^2$ . Best predictor:  $f^*(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}]$ .

Reason: for fixed  $\mathbf{x}$ ,  $\mathbb{E}[(Y - c)^2 \mid \mathbf{X} = \mathbf{x}]$  is smallest at  $c = \mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}]$ .

### Classification: 0-1 loss

$L(y, f(\mathbf{x})) = \begin{cases} 1, & y \neq f(\mathbf{x}) \\ 0, & y = f(\mathbf{x}) \end{cases}$ . Best classifier:  $f^*(\mathbf{x}) = \arg \max_{y \in \mathcal{Y}} p(y \mid \mathbf{x})$ .

Reason: given  $\mathbf{x}$ , the error probability is  $1 - p(f(\mathbf{x}) \mid \mathbf{x})$ , minimized by picking the most likely class.

## Your turn

Joint table  $p(x, y)$

|         | $y = 0$ | $y = 1$ | $y = 2$ |
|---------|---------|---------|---------|
| $x = 0$ | 0.3     | 0.1     | 0.1     |
| $x = 1$ | 0.05    | 0.15    | 0.3     |

Find the best regression predictor and the best classifier, for each  $x$ .

Answer

## Your turn

Joint table  $p(x, y)$

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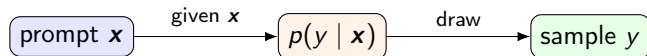
Answer

$p(y \mid x = 0) = (0.6, 0.2, 0.2)$ :  $\mathbb{E}[Y \mid X = 0] = 0.6$ , best class 0.

$p(y \mid x = 1) = (0.1, 0.3, 0.6)$ :  $\mathbb{E}[Y \mid X = 1] = 1.5$ , best class 2.

1.5 is not a value  $Y$  can take. Regression and classification ask different questions of the same joint.

Knowing the joint buys one more thing: we can **generate** from it.



- Sample a fresh pair  $(\mathbf{X}, Y)$  from  $p(\mathbf{x}, y)$ .
- Given  $\mathbf{x}$ , sample  $Y$  from  $p(y | \mathbf{x})$ .

For a language model: sample a response  $y$  given a prompt  $\mathbf{x}$ .

**Later:** Sep 1 builds a language model exactly this way.

# The three tasks side by side

| Task          | Unknown            | Distribution known?        | Uncertainty   | Where                      |
|---------------|--------------------|----------------------------|---------------|----------------------------|
| Reasoning     | nothing            | yes                        | noise         | Sec. 1.1                   |
| Inference     | a parameter(s)     | form known, parameters not | finite sample | estimation chapters        |
| Supervised ML | $p(\mathbf{x}, y)$ | no                         | both          | regression, classification |

Two consistently important questions: what is unknown and where does the uncertainty come from.

- Inference: a parameter, fixed but unknown (frequentist) or random (Bayesian); uncertainty from finite sample size.
- Supervised learning: the joint  $p(\mathbf{x}, y)$  is unknown; uncertainty from noise and finite sample size.
- If the joint were known, the best predictor and classifier follow directly; knowing it also lets us generate.
- With only data, two routes: minimize empirical risk, or estimate the joint.

**Next:** model complexity and language models, Sec. 1.3.